The electric field between the plates and outside the plates is found using Gauss law. The magnitude of the electric field due to an infinite charged plane sheet is $\frac{\sigma}{2\varepsilon_0}$ and it points perpendicularly outward if $\sigma > 0$ and points inward if $\sigma < 0$.

At the points P_2 and P_3, the electric field due to both plates are equal in magnitude and opposite in direction (Figure 1.41). As a result, electric field at a point outside the plates is zero. But inside the plate, electric fields are in same direction i.e., towards the right, the total electric field at a point P_1

$$E_{inside} = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0} \qquad (1.72)$$

The direction of the electric field inside the plates is directed from positively charged plate to negatively charged plate and is uniform everywhere inside the plate.

(iv) Electric field due to a uniformly charged spherical shell

Consider a uniformly charged spherical shell of radius R and total charge Q as shown in Figure 1.42. The electric field at points outside and inside the sphere is found using Gauss law.

Case (a) At a point outside the shell (r > R)

Let us choose a point P outside the shell at a distance r from the center as shown in Figure 1.42 (a). The charge is uniformly distributed on the surface of the sphere (spherical symmetry). Hence the electric field must point radially outward if Q > 0 and point radially inward if Q < 0. So we choose a spherical Gaussian surface of radius r is chosen and the total charge enclosed by this Gaussian surface is Q. Applying Gauss law

$$\oint_{\substack{Gaussian\\surface}} \vec{E} \cdot d\vec{A} = \frac{Q}{\varepsilon_0} \qquad (1.73)$$

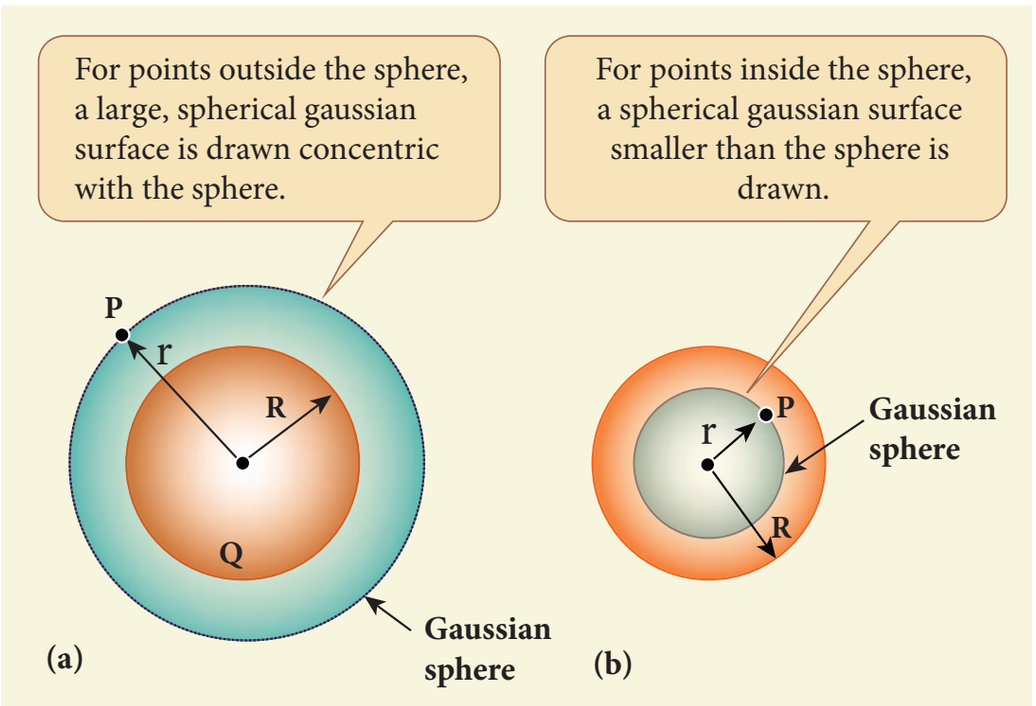


Figure 1.42 The electric field due to a charged spherical shell

UNIT-1(XII-Physics_Vol-1).indd 46 04-03-2019 10:26:14